



Regulatory Instability, Business Process Management Technology, and BPM Skill Configurations

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Abstract. This paper investigates how firms configure their business process management efforts in different industries. We generate a business process management (BPM) skills taxonomy through the computational linguistic analysis of job ads from [Monster.com](https://www.monster.com). We apply the taxonomy to [LinkedIn.com](https://www.linkedin.com) resumes of professionals employed at retailer Walmart, pharmaceutical company Pfizer, and investment bank Goldman Sachs. We find that Walmart and Pfizer distribute change- and operations-related BPM skills among the same roles whereas Goldman Sachs distributes both kinds of skills among more separate roles. This separation reflects a trilateral configuration where line managers and analysts focus on operational BPM tasks related to running processes while change-related tasks are covered by project managers. At Walmart and Pfizer the tasks of the BPM project manager are shared among managers and analysts, reflecting a bilateral configuration. Comparing each firm's regulatory environments and BPM technology capabilities, we conjecture that the organizational configuration pattern is influenced by a firm's ability to reliably automate business processes, since this affects how much attention line managers and analysts have to spend on monitoring processes and on reconciling issues and exceptions. This attention could otherwise be spent on regulatory-imposed process change efforts. This configural logic suggests a reconfiguration of BPM professionals towards a bilateral configuration when an organization transforms its business with digital technology, because the focus of such efforts includes process and decision automation.

Keywords: BPM skills · BPM taxonomy · BPM professionals · BPM function

1 Introduction

Regulatory interventions are a frequent source of organizational change in industries such as Pharmaceuticals or Financial Services, whereas other industries such as Retail or Transportation are subject to more measured changes of the regulatory regime. By regulation we mean authoritative operating rules accompanied by some formal governance mechanisms that promote rule compliance and sanction non-compliance and misconduct [2]. Governments typically impose regulation on organizations to increase market stability and transparency by permitting, directing, constraining their operations. When regulations change, organizations need to adapt their policies and

procedures, and hence they require business process management (BPM) professionals and a BPM function that can absorb the regulatory instability of their regulatory regime, akin to Ashby's [1] law of requisite variety. Which actions organizations take to match the complexity of their BPM function to that of their regulatory regime is thus an important question for regulators and those affected by regulation alike.

A regulatory regime is unstable when it comprises multiple supervisory authorities that collectively impose rules that are complex and frequently changing. A stable regulatory regime can be characterized by a limited number of authorities and a continuance of rules over time. The instability is different from a regulatory shock, i.e., sudden and extensive changes in the operating constraints [12]. *Regulatory shocks* are typically less predictable and more harmful to the business practices of affected organizations, either because they occur unanticipated (e.g., the imposition of tariffs in a trade conflict) or organizations delay the implementation of substantial rule changes (e.g., new privacy laws). In contrast, *regulatory instability* is a second-order measure of the delta in month-over-month, year-over-year changes in the rule book imposed by the regulatory regime that directs and constrains how business processes can be performed, making it a more predictable variable to which organizations can adapt and attend to.

Individuals are cognitively bounded with regard to how much information they can process at a time, and hence organizations need to configure their professionals to distribute their attention to salient issues and their solutions [22]. When organizations face a regulatory shock such as the introduction of the U.S. Sarbanes-Oxley Act, they often increase and/or complement their business and technology staff through new hires and audit and consulting engagements to ensure regulatory adaptation, and invest in regulatory technology to automate compliance monitoring and control [14, 16]. Unlike a shock, however, an unstable regulatory regime should require continual attention, and it stands to question whether affected organizations simply require more professionals than those operating under a stable regime or whether these organizations make more permanent changes to the configuration of their BPM function to reflect the different tasks performed by different professionals on a daily basis.

The skills of the professionals involved in BPM and their configurations can serve as a suitable proxy for the BPM function overall because organizations frequently practice BPM as a method to analyze, design, implement, monitor, and control their operations, treating processes as the socio-technical change objects [4]. When organizations practice BPM, they assume that processes have an ostensive and a performative aspect [9]. The ostensive aspect describes the process as a model and executable script that specifies the logical workflow design and implementation. The performative aspect relates to the actions performed by human and algorithmic system participants. The skills of BPM professionals constrain the change-related (ostensive) and operations-related (performative) BPM tasks they could possibly perform [21], and their collective configuration provides an upper bound of the ordinary and dynamic capabilities of what the BPM function can achieve in the context of an organization [18, 24].

In this paper, we ask *how the BPM function can be affected by regulatory instability, how such demands translate into different configurations of BPM professionals, and whether BPM technology can mediate between regulatory instability and these configurations*. By *BPM technology* we mean algorithms and computer systems that

automate operational tasks and processes, substituting human performance without causing exceptions that require human attention [26]. First, we empirically generate a skills taxonomy from job ads. Second, we apply the taxonomy to public resumes of professionals employed at three organizations from differently regulated industries. Specifically, we look at the BPM skills configurations at retailer Walmart, pharmaceutical company Pfizer, and investment bank Goldman Sachs, exposing the skill configurations of their BPM functions. We interpret the configurations considering the relative differences in the stability of each company's regulatory regime and BPM technology capability.

We find that while the fundamental skills within the BPM functions of Walmart, Pfizer, and Goldman Sachs are similar, their configurations differ. Change- and operations-related BPM skills are distributed among line managers and analysts at unstable-regulated Pfizer. The configuration mirrors that of stable-regulated Walmart, reflecting a bilateral BPM configuration where both roles possess the necessary skills to collectively operate and adapt processes. The skills to operate and adapt processes are more distributed at Goldman Sachs which relies on project managers to complement its line managers and analysts in process change efforts. Goldman Sachs allows its line managers and analysts to attend more to operating business processes, reflecting a trilateral BPM configuration. Comparing the BPM technologies of Pfizer and Goldman Sachs, we conjecture that the BPM configuration also depends on the capacity to automate processes affected by unstable regulation. A sophisticated BPM technology capability can allow line managers and analysts to shift their attention to change-related BPM tasks rather than compliance and escalation-related operational tasks, rendering the project manager redundant when performative process exceptions are caught and resolved by BPM technology as specified in ostensive models and scripts.

2 Regulatory Instability and BPM Professionals

Regulation affects an organization's processes and systems. Organizations that produce similar products or services are historically assigned to the same industry [8], and hence are regulated by the same regulatory regime. Although the designs of their processes often differ, regulation manifests as controls that must be implemented within existing processes (e.g., margin requirements of an investment bank's client borrowing liquidity to trade securities), or that formulate new processes (e.g., which party reports a trade), or that must not be performed (e.g., the trading desk must not engage in proprietary trading). Not all regulation permits and limits the product and service portfolio of organizations, but it may affect internal operating policies and procedures. The stability of the regulatory regime therefore not only varies between organizations from different industries but also between processes and units within the same organization.

As for the United States, the amount of regulation published over time through the Office of the Federal Register can serve as a suitable proxy of an industry's regulatory stability.¹ The number of regulators and rules which organizations must monitor,

¹ A list of publications released can be provided by the authors on request.

comply with, and adapt to depend on the size of their product and service portfolio and on their operational footprint because regions and countries can have their own authorities and standards. As for the European Union, it is safe to say that the stability of regulatory regimes is comparable to their U.S. peers. In relative terms, organizations in Consumer Goods and Financial Services are regulated by a more unstable regime, while organization in Retail experience a more stable regime.

Like other individuals, BPM professionals have a finite absorptive capacity to attend to all information. According to the attention-based view, organizations need to specify roles assumed by individuals and their reporting relationships to distribute time and effort to issues that require attention [22]. By *attention* we mean the noticing, enticing, and focusing of time and effort on issues (i.e., problems, threats, and opportunities) and their solutions facing the organization [22]. An unstable regulatory regime is a significant concern for affected organizations, and their challenge is to configure skills into roles so that everyone can effectively perform a finite set of tasks, attending to the subset of information relevant to their tasks. Collectively, the configuration should maintain the capacity to monitor and ensure day-to-day operational compliance and at the same time ensure a timely response to rule volatility imposed by the unstable regime.

The skills required of individuals to perform BPM tasks relevant to their role are commonly outlined in job descriptions [20]. To maintain a BPM capability, organizations need to articulate the skills they deem desirable using some formal or technical language and assign them to roles. BPM roles require operational knowledge and thus sit at middle and lower managerial ranks. We therefore expect organizations independent of their industry to also publicly hire to fill vacancies.

3 Method

We select a method to computationally explore and extract skills from job descriptions that does not rely on a pre-specified classification schema. The categories of such a taxonomy (in our case skills) are unknown because we cannot specify ex-ante which skills organizations seek in the context of their BPM practice and what words they use to articulate their needs. We acknowledge that O*Net provides a generic taxonomy for skills across different professions, but we found that this taxonomy would not allow us to discriminate between managerial and technical professional roles that are characteristic of BPM [17].

3.1 Datasets

Job Ads. We obtained job descriptions from job ads because they describe the perceived deficiencies of organizations in their staffing to maintain specific capabilities [23]. We downloaded the complete set of job ads that contained *business process*, *BPM*, *process improvement*, *process innovation*, or *process change* from [Monster.com](https://www.monster.com), resulting in 45,484 job ads published in the U.S. and U.K. between January 2015 and 2016. We removed 20,121 ads related to non-permanent and heavy-industry production

occupations, leaving 25,363 job ads. We focused on both countries because they share a similar regulatory regime and language. We randomly sampled 1,000 job ads from Retail (NAICS: 44), Consumer Goods (31), and Financial Services (52), respectively, in order to normalize the amount of text representing any particular sector. Without this step, sectors with more job ads could have biased the generated skills taxonomy. We sampled job ads published by organizations for which we could confirm their industry affiliation through Dun & Bradstreet.

Resumes. Job ads provide a partial window into organizations because they do not expose the skills and roles which they already have. We used resumes to obtain a more complete understanding of the BPM professionals at three theoretically sampled organizations. From [LinkedIn.com](https://www.linkedin.com) we downloaded resumes of individuals who self-declared (as of September 2017) to work at retailer Walmart, pharma Pfizer, or investment bank Goldman Sachs. We only included those who used *business process*, *BPM*, *process improvement*, *process innovation*, or *process change* to describe their current full-time occupation. Again, we randomly sampled 1,000 resumes for each organization.

The three organizations are all large-scale in their employee size and among the most professionally-managed ones in their respective industry. From interviews with senior managers of each company, we learned that Goldman Sachs' process automation capability was constrained by that of their counterparties. Pfizer did not experience this degree of interdependence, being less dependent on external parties to complete a business process. Unlike Pfizer, Walmart and Goldman are service organizations but due to the nature of their transaction-like business operations, they have quasi-manufacturing processes.

3.2 Procedure

We generate a BPM skills taxonomy and map the skills of the professionals at Walmart, Pfizer, and Goldman Sachs into the taxonomy. First, we explore latent topics (in our case skills) across all job descriptions. Latent topics are word groups associated with a semantic context. In job ads, the semantic context can be thought of as skills, assuming that organizations can use different words to advertise the same skill. Second, we infer a skill set over each resume and cluster the skill sets for each firm, thereby exposing BPM configurations.

Modeling Skills. We use a topic model widely accepted in the computational linguistics literature [15], which is Latent Dirichlet Allocation (LDA) [5]. LDA is a generative model for classifying documents into multiple topics that are latent in their structure. A topic is a probability distribution over a lexicon. Topics are shared by all documents and the lexicon is shared among all topics. Words that have a high probability within a topic tend to co-occur across documents. LDA infers the classification scheme in form of word-topic and topic-document distributions.

The number of topics expected in the documents must be specified prior to model inference, and the optimal number can be identified by comparing their validity. We

use mean topic coherence as an indicator of internal validity, which measures how accurately high-probability words of a topic do co-occur across documents [6]. Perplexity measures external validity, the capacity of the inferred word-topic distributions to capture word co-occurrences in previously unseen documents [5]. We run three-fold cross-validation to train and test each model, obtaining a more robust score. Both mean topic coherence and perplexity are technical scores and their interpretation should not go beyond model comparison.

Mapping Skills and Roles. We infer a topic-document distribution over an imaginary document composed of a resume's current job title, description, and skills using the topic model which we identify as optimal in the previous step. To explore variation in the skill configurations, we group the topic-document distributions and run separate cluster analyses for each firm. We cluster skill sets based on their pairwise Hellinger distances using a bottom-up algorithm (the Ward method) that aims to minimize the total within-cluster variance at each level. The emerging clusters can be thought of as distinct roles, i.e., skill sets. To interpret the clusters in a particular layer, we look at the most-representative resumes which we define as those whose topic-document distributions are closest to the mean distribution, i.e., the topic-role distribution, of all resumes assigned to a particular cluster. In other words, we look at the cluster centroids.

3.3 Analysis

We performed a number of text reduction steps to the job ads [15]. The outcome was a lexicon and sentence vectors counting the frequency with which each word occurred in a sentence. Not all information provided in job ads refer to the job description [23], and hence we cleansed them.

Removing Duplicate Job Ads. We removed duplicates based on the cosine distance of their aggregated sentence vectors. The .10-threshold yielded the best F-measure (.90) in detecting duplicates as classified by two researchers (Cohen's $\kappa = .78$). The threshold was conservative, given that the average job ad contained 93 distinct words (s.d. = 46). We identified 7,860 job ads as being duplicates, leaving 17,503.

Removing Noise Sentences. We applied LDA over the sentences to remove those that obviously did not address skills but contextual information. The idea was to interpret each sentence as a distribution of domain topics and noise topics, and to remove sentences from a job ad that exhibit a high probability of being associated with a noise topic. The 40-topic model provided the best balance between internal and external validity, separating distinct themes into different topics. We treated a topic as noise when it addressed a theme that was obviously not domain-related ($\kappa = .93$). A .50-threshold yielded the best F-measure (.82, $\kappa = .74$) in detecting noise sentences, resulting into a removal of 77,700 sentences (19.6%).

Modeling Skills. We aggregated the remaining sentence vectors for each job ad and reran LDA. The 24-topics model generated substantive topics and provided the best

balance between internal and external validity. Two researchers independently labeled each topic and synthesized final labels with a third researcher.²

Mapping Skills and Roles. We selected the 24-topic model to infer a topic-document distribution over each resume's relevant sections which we identified by their proper HTML tags. We grouped the resumes for each firm and evaluated the cluster solutions. The best solution was identified by a distinct bent when being plotted and validated using the gap statistic which measures the goodness of a solution relative to a randomly sampled solution of the same number of clusters [25]. Both indicators suggested the five-cluster solution for Walmart and the six-cluster solution for Pfizer and Goldman Sachs, which produced simple models that exposed the fundamental structural differences between their respective BPM skill configurations. Two researchers independently inferred cluster labels from the skill configurations and job titles of the representative resumes and synthesized them with a third researcher.

4 Findings

We interpret the BPM skills as being related to either change- or operations-related tasks. Change-related BPM skills help to intentionally change the design, models, organizational structures, and software code of a business process (i.e., ostensive aspects [9]) and to translate such changes into operations (i.e., performative aspects). Operations-related BPM skills support individuals to align the actual operations of the process with its intended design to stabilize and standardize them. Together, both types of skills provide the basis for adapting and operating processes and systems.

Figures 1, 2, and 3 illustrate the skill configurations at Walmart, Pfizer, and Goldman Sachs. The dark crosses and circles represent the change- and operations-related skills ($\kappa = .94$). Their relative positioning on the map reflects the similarity of their underlying word-topic distributions generated by our LDA. To compare differences in the configurations, the opaqueness of a curve that links a skill with a cluster center is a function of whether the skill receives a cluster relevance that is within the 90th, 80th, 70th, or 60th percentile of skill attentions across all clusters.

4.1 BPM Professionals at Walmart

The professionals within Walmart's BPM function suggest a configuration into three line manager and two analyst roles (see Fig. 1). The *front-office manager* aggregates professionals skilled to perform and administer customer-facing functions such as marketing and sales. The *back-office manager* bundles those who work in enterprise- and partner-facing functions, e.g., finance and accounting. The *operations manager* comprises supply chain and logistics professionals. All three manager roles possess skills related with operations-related BPM, however, some change-related skills appear to be boundary-spanning in the sense that they integrate the roles: *business strategy*, *product development*, *business transformation*, and *change management* are shared

² A list of high-probability words per topic can be provided by the authors on request.

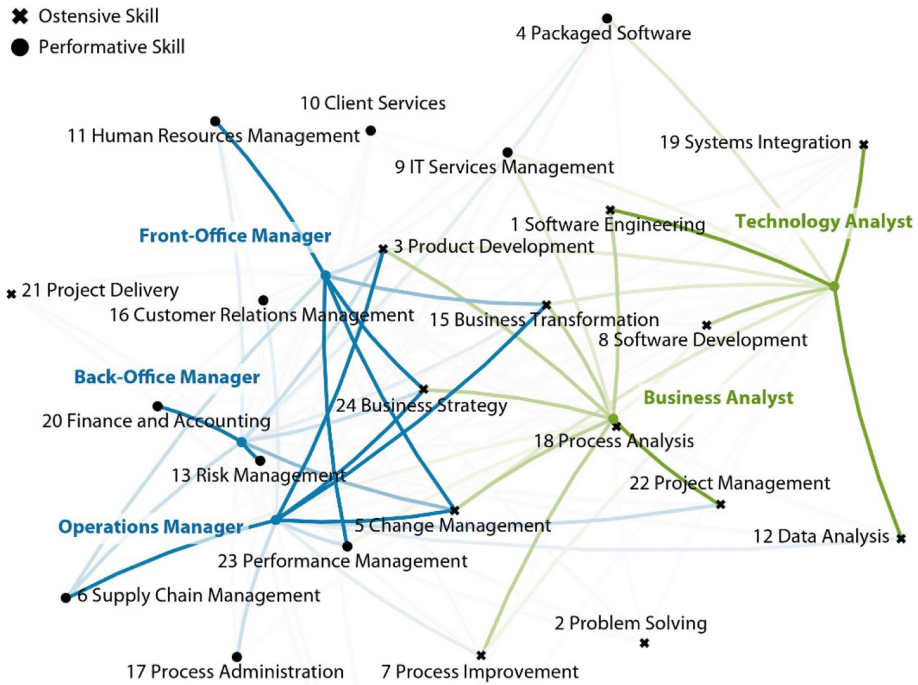


Fig. 1. BPM configuration at Walmart

change-related skills. These managers are typically responsible for the bottom-up championing and the top-down implementation of strategic programs, and hence they need the people skills necessary to execute and deliver operational changes, reflected in the *performance management* skill of *front-office* and *operations managers*. Walmart's back-office operations are highly-automated with their SAP systems, and hence performance management is less of a concern for *back-office managers*.

The *business analyst* role links to typical process project-related skills such as *project management*, *process analysis*, and *software engineering*, whereas professionals aggregated under the *technology analyst* role are skilled in *software development* and *systems integration* but also *data analysis*. Both roles link mostly to change-related BPM skills with the *business analyst* occupying a central position within the BPM configuration as an interface between business and technology. This configuration mirrors the common understanding of analyst role in the literature as a boundary spanner and knowledge broker. This role appears to be more assigned to a process on a project basis in order for its optimization or change.

The skills of *technology analysts* reflect Walmart's IT strategy to have a commercial ERP system for the standardization of non-differentiating processes, while engineering merchandizing and supply chain systems more on their own as a measure to better integrate e-commerce and department-store operations. The responsibilities for new software developments reside in Walmart's digital business unit Global

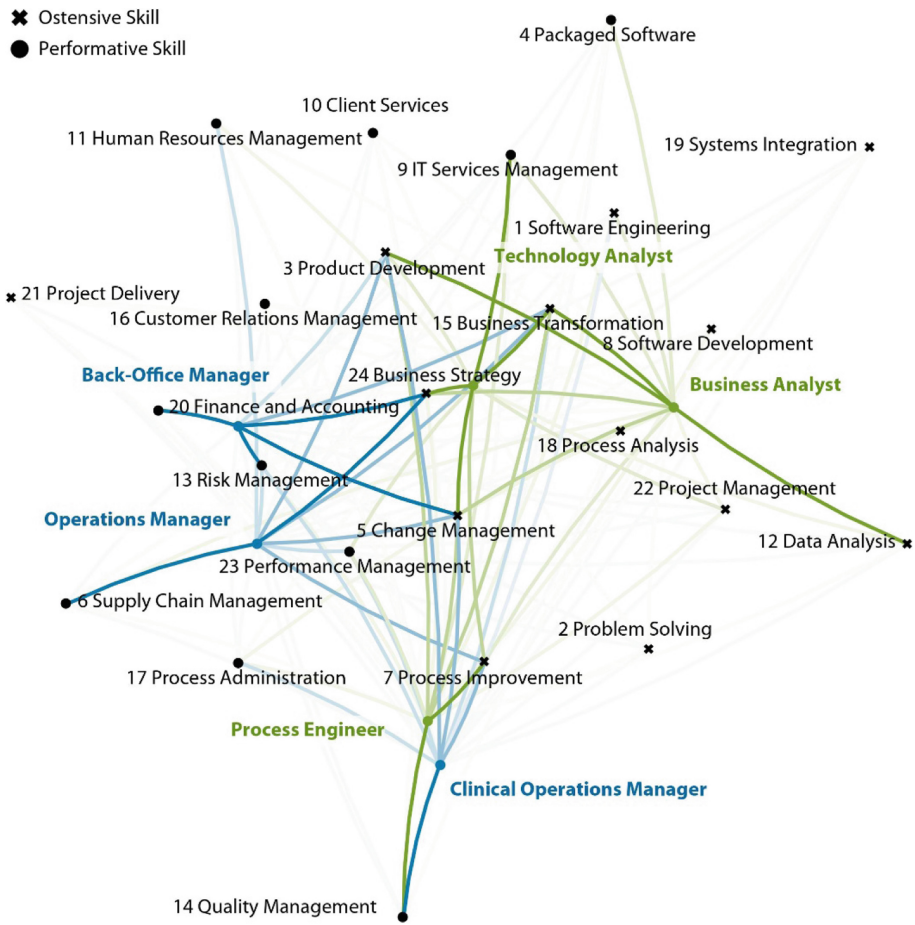


Fig. 2. BPM configuration at Pfizer

eCommerce, and hence it is plausible that Walmart’s *technology analysts* are mostly skilled in the integration of backbone application systems.

4.2 BPM Professionals at Pfizer

The professionals within Pfizer’s BPM function are grouped into three manager and three analyst roles (see Fig. 2). The *back-office manager* and *operations manager* comprise professionals with skills similar to their peers at Walmart which makes sense due to the similarity of their business models of selling physical products. Pfizer sells its products to commercial businesses rather than end consumers, which can explain the missing *front-office manager* role because out-bound processes are under the auspices of operations managers. Instead, a *clinical operations manager* role bundles the professionals skilled to monitor and control the new drug and medicine development processes. Professionals in this role possess operations-related BPM skills related to

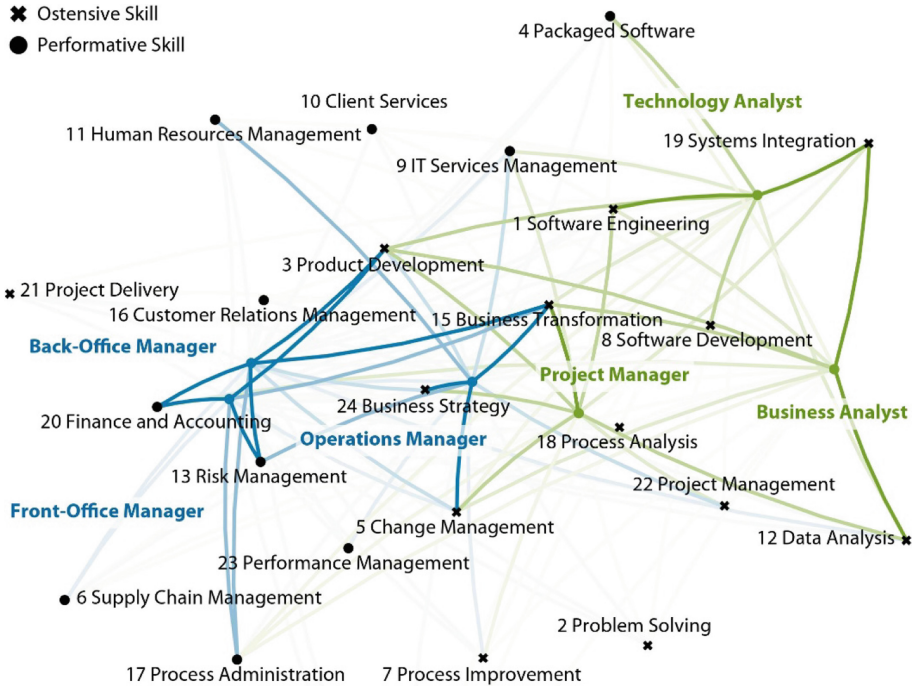


Fig. 3. BPM configuration at Goldman Sachs

process administration, risk management, and quality management, and change-related skills such as *product development*, *process improvement*, and *change management*.

Pfizer's manufacturing operations add the *process engineer* role that comprises professionals with change-related skills relevant for the design of physical production processes. These professionals have similar skills as *clinical operations managers*, however, *process engineers* are those who can engineer reliable manufacturing processes, according to their resumes. *Process engineers* are skilled to analyze, design, and improve the high-volume processes whose performances are highly automated within the firm. The *clinical operations manager* monitors and controls processes such as clinical trials which involve different actors from within and outside of the firm and which are less standardizable in their performance. The *business analyst* role summarizes skills akin to its peer at Walmart. Pfizer's *technology analysts* are more strategic-oriented towards change, as reflected in the links to *business strategy*, *product development*, *business transformation*, and *change management*. Neither the *business analyst* nor the *technology analyst* exposes software development and engineering skills, suggesting a skills chasm between the firm's BPM and IT function.

4.3 BPM Professionals at Goldman Sachs

The professionals within Goldman Sachs' BPM function are grouped into four manager and two analyst roles (see Fig. 3). The *front-office* and *back-office managers* comprise

professionals with skills mostly related to operations-related BPM such as *risk management*, *process administration*, and *finance and accounting*, however, the development of new innovative financial products is also part of their skill sets. This suggests that these professionals see their role more focused on identifying, monitoring and controlling (new ways of) revenue-generating client interactions and transactions in alignment with the firm's risk appetite. The *operations manager* bundles the professionals skilled in change-related BPM, according to the links to *business strategy*, *product development*, *business transformation*, and *change management*. *Risk management* and *IT services management* being operations-related skills seem to be also their concern.

The emphasis of skills of Goldman Sachs' managers on emphasizing risk, compliance, and administration compared with a relatively technical orientation of their *business analysts* arguably necessitates the existence of the *project manager* role that comprises professionals with design and change skills to coordinate between business and technology. These professionals possess a skill set comparable to Walmart's business analysts.

5 Discussion

We asked how organizations configure their BPM professionals to attend to the changes caused by regulation. Our three cases suggest that organizations whose processes are regulated by an unstable regime separate change- from operations-related BPM tasks if the processes require extensive manual monitoring and control by managers and analysts. If operations-related BPM tasks are automated by BPM technology, less manual operational attention is needed. A higher BPM technology capability could allow end-to-end workflow automation with minimum exceptions, substituting human by algorithmic attention. Eliminating humans from a process can reduce variation between intended and actual process operations because processes are executed as specified by the model and executable script. This allows managers and analysts to attend more to change-related BPM tasks and regulatory adaptation. This configuration is reflected in the BPM professionals of an organization that is regulated by a stable regime. We thus distinguish between a *bilateral* and a *trilateral BPM configuration* and conjecture that the configuration of an organization operating under an unstable regulatory regime also depends on its BPM technology capability.

5.1 Bilateral and Trilateral BPM Configuration

The intended (ostensive) and actual (performative) operation of a process are in constant friction. That means the performative process can become the ostensive process when being performed repeatedly in deviation from its intended design, and the ostensive process may change even though this change does not materialize in the actual performance of the process [9]. Regulation affects an organization's ostensive processes, followed by the performative processes. That said, regulatory compliance is assessed against the performative processes, not the ostensive ones.

The *ostensive misfit* between external regulatory requirements and internal ostensive processes and the *performative misfit* between ostensive and performative processes impose pressures on an organization and demand attention. Both ostensive and performative misfits represent “exceptions,” and BPM professionals have finite attention to address and solve all exceptions given finite time. Depending on the frequency of regulatory changes triggering ostensive misfits and the provenance of performative misfits, we propose that organizations rely either on a bilateral or trilateral configuration of their BPM professionals to maintain the necessary absorptive capacity to attend to these exceptions and their resolution (see Fig. 4).

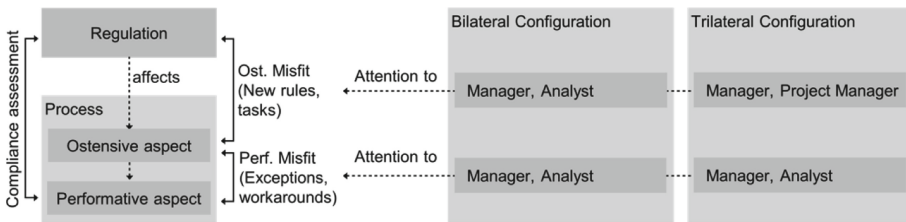


Fig. 4. Bilateral and trilateral BPM configuration

If the regulatory regime is stable, then exceptions are mostly caused by performative misfits. A bilateral BPM configuration allows to align the performance of processes to their intended designs, and to execute and implement intended changes because professionals combine the necessary BPM skills for design and execution. Performative misfits are either caused by individuals or technology performing process activities. Managers can coordinate and solve human misalignments whereas algorithmic misalignments are solved by analysts. Managers monitor and control everyday process operations to maintain performative fit, applying methods such as Lean Management to alter ostensive processes and Six Sigma to translate such changes into measurable performative improvements. These managers have the skills to identify performance problems and their root causes, and take action for their resolution. Business and technology analysts address algorithmic misalignments, and resolving such misfits is an ostensive rather than performative problem because it requires changes to executable scripts. The bilateral interactions between the professionals required to change ostensive and performative processes are characteristic of the bilateral BPM configuration.

Regulation is published as a complex web of textual non-machine-readable documents and are often written with a significant amount of ambiguity. Implementing new regulations is not a straight-forward task. It imposes significant amounts of sense-making time on human individuals. This sense-making involves understanding the documents and their relations to others, assessing implications on processes, and searching for an optimal way to implement requirements and change policies and procedures. This adaptation demands special attention [26].

Martinez-Moyano et al. [19] explain how agency problems in investment banks such as Goldman Sachs cause profit maximization pressures that run against compliance, increasing rule violations that in the absence of intervention derail the business. Pfizer faces similar tensions between maximizing output and ensuring regulatory compliance of its production processes. Its higher BPM technology capability though requires managers and analysts to spend less of their time with monitoring and control the compliancy of their operations, reflecting a bilateral BPM configuration. These processes have a higher straight-through processing rate to benefit from scale economies, allowing managers and analysts focus more on engineering rather than administration. It is therefore not surprising that these professionals are more engineers than classical analysts, indicating the comprehensive tasks performed by these professionals.

If the regulatory regime is unstable *and* processes *can* be reliably automated by BPM technology to cause minimum performative misfit that requires human intervention, ostensive misfits caused by differences between an organization's regulatory requirements and ostensive processes become the major source requiring attention. Under such a condition, managers and analysts can focus on regulatory realignment because BPM systems control and perform processes and activities. Ostensive changes to executable scripts directly translate into their automated performance.

Unlike Pharmaceuticals, Financial Services is a heavily vertically-disintegrated industry with by many specialized firms buying and selling financial products both electronically and over the counter. Goldman Sachs processed more than one million trades per day with a straight-through processing rate of about 96% at the time of our data collection, with a significant amount of time of their operations managers and analysts being consumed by reconciling errors and interruptions. While Pfizer has more control of its unstably-regulated processes because it can buffer them from upstream and downstream partners, the capability of financial services providers to perform with minimal exceptions also depends on the technology capability of their counterparties because financial services production and delivery processes are heavily inter-organizational. The capability algorithmically buffer internal operations from, and reconcile, counterparty-caused exceptions depends on a firm's ability to develop automatable ostensive scripts that can algorithmically catch and handle such issues. The complexity in market interactions makes such engineering efforts inherently difficult.

If the regulatory regime is unstable *and* performative processes *cannot* be reliably automated because BPM technology has a lower capability, the organization simultaneously faces pressures from ostensive and performative misfits. Under such a condition, we propose that the organization relies on a trilateral BPM configuration because managers and analysts are more concerned with reconciling and resolving performative misfits to ensure that processes achieve their desired outcome. Therefore, a project manager addresses ostensive misfits, being in charge of redesign and adaptation. The trilateral interactions between the professionals required to change ostensive and performative processes allow an organization to maintain the requisite absorptive capacity to cope with an unstable regulatory regime when operational attention cannot be shifted to BPM technology.

Not all processes of an unstable-regulated organization experience regulatory instability. For example, regulatory requirements of typical support processes such as

human resources and payroll are often stable. Different unstable-regulated processes typically also have a different BPM technology capability. We conjecture that organizations change such processes by applying a bilateral approach, coordinated by analysts rather than project managers because such processes require less attention to ostensive misfits. The bilateral and trilateral BPM configurations are not necessarily mutually-exclusive but can coexist within the same organization. Rather, these organizations situationally switch between a bilateral and trilateral approach to process management within their trilateral BPM configuration. We find support for this conjecture in Goldman Sachs' *analyst* and *project manager* clusters, which absorb process engineers in their periphery. These engineers have skills akin to Pfizer's process engineers – for digital rather than physical processes.

5.2 Implications

While our analysis is motivated by the bottom-up clustering and aggregation of skills of human individuals into collective skill configurations at three industry-leading firms, in practice it is more plausible that these organizations take a decompositional approach by assessing their regulatory regime and BPM technology capability first, and by specifying the roles and their required skills accordingly. Regardless of whether such specifications are based on macro-to-micro decomposition principles or micro-to-macro aggregation principles, or a mix of both, our cross-sectional analysis cuts levels of organizational analysis. We show how macro-level effects about regulatory instability and technology can relate with micro-level skill configurations of an organization's BPM practice. These configurations represent a human resources-based perspective on the microfoundations of an organization's ambidextrous process management capability that addresses the ostensive and performative issues that require attention [20]. Arguably because of a shortage of analysis techniques and datasets, the linkage between microfoundations (individual, skills) and macrofoundations (organization, capability) is mostly a theoretical debate [3, 10]. Our contributions here are method-wise, and our approach to uncover skills and configurations can help other academics better understand the linkages between individual and organizational aspects affecting human resources and organizational performance both within and outside of BPM [7].

On the practical side, our discovered configurations can serve as a basis for professionals in charge of a BPM practice better assess their current BPM target operating model, linking roles, skills, and capability. Digital and information technologies reduce vertical layers of management hierarchy [11, 13]. However, BPM technology may also alter the horizontal structuring of managers when processes become digitized and their operations automated. BPM technology capability creates options for Chief Process Officers to rethink the skill sets required of their staff to better enable and support the digital transformation of their organization over time. As organizations continue to automate their routine and non-routine tasks with artificial intelligence and machine learning, robotic process automation, cognitive agents, and distributed ledger technology, among other technologies, these digital transformations turn operations-related BPM tasks inherently into engineering problems. Chief Process Officers should keep an eye on their BPM technology capability and put in place plans for the enterprise-wide education and reskilling professionals involved in the operational management and

administration of processes to allow for a shift from a potential trilateral configuration to a bilateral one. We speculate that digitally-transformed organizations can allow their managers and analysts to attend more to the innovation of minimum viable products and services. However, our analysis does not consider performance measures of both configurations, and hence future research should try to link both configurations with organizational performance implications.

5.3 Limitations

While our study provides important insights into the skills and configurations of BPM professionals, our findings are subject to a limitation that relates to the nature of job ads and self-reported resumes as a research dataset. For example, an organization may announce a presumed vacancy less with the motivation to fill this role but to be recognized as a legitimate business partner for (potential) customers. Unstable-regulated industries may be more likely to write about governance, risk, and compliance in their job ads than organizations operating under a more stable regulatory regime, ultimately paying less attention to other skills and tasks relevant to the role.

We tried to mitigate this concern by generating a common skills taxonomy across the Retail, Consumer Goods, and Financial Services industries rather than an industry-specific skills taxonomy, and by mapping the resumes into the common taxonomy. Similar concerns may exist for the skills and job descriptions reported in resumes. Professionals certainly use [LinkedIn.com](https://www.linkedin.com) to connect and advertise themselves to increase their visibility and attractiveness for other employers, and hence they will likely select words in their resumes that add to their employability. We addressed this potential self-reporting bias methodologically by selecting a randomly-sampled and sufficiently large amount of resumes, and by averaging the skills reported in each research, looking at the cluster centroids rather than peripheries. The findings provide insights into the *means* and not ends of the BPM configurations of three theoretically-sampled organizations. Future research should test the BPM configurations and conjectures to a broader set of organizations, industries, and aim to establish a link between BPM skill configurations and performance implications of these organizations.

6 Conclusion

An organization's BPM technology can play an important mediating function between its regulatory regime and the configuration of its BPM professionals. Investment bank Goldman Sachs is an example of how a lower process automation capability can be associated with a trilateral BPM configuration that is used to maintain absorptive capacity to attend to the ostensive and performative misfits caused by regulation and operation. Pfizer is an organization that manages to maintain a higher automation capability within an unstable regulatory context and appears to be able to transfer performative BPM tasks from its professionals to the technology, reflecting a bilateral BPM configuration. This configuration mirrors that of an organization that operates under a stable regulatory regime, exemplified by our analysis of Walmart.

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